



Constraints on the Metagalactic Hydrogen Ionization Rate from the Lyman- α Forest Opacity

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Overview

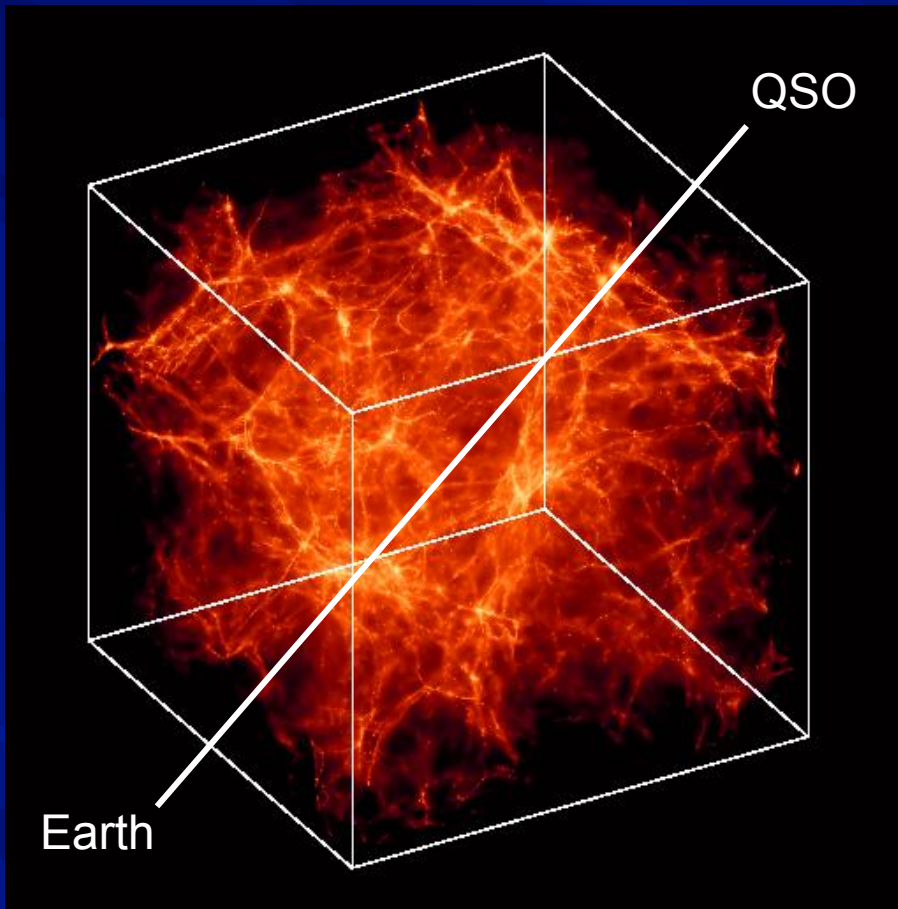
Motivation:

- What is the intensity and spectral shape of the UV background?
- Constrain the sources responsible for reionizing the IGM
- Probe the thermal history the IGM – implications for epoch of reionization

Probes of the UV background intensity:

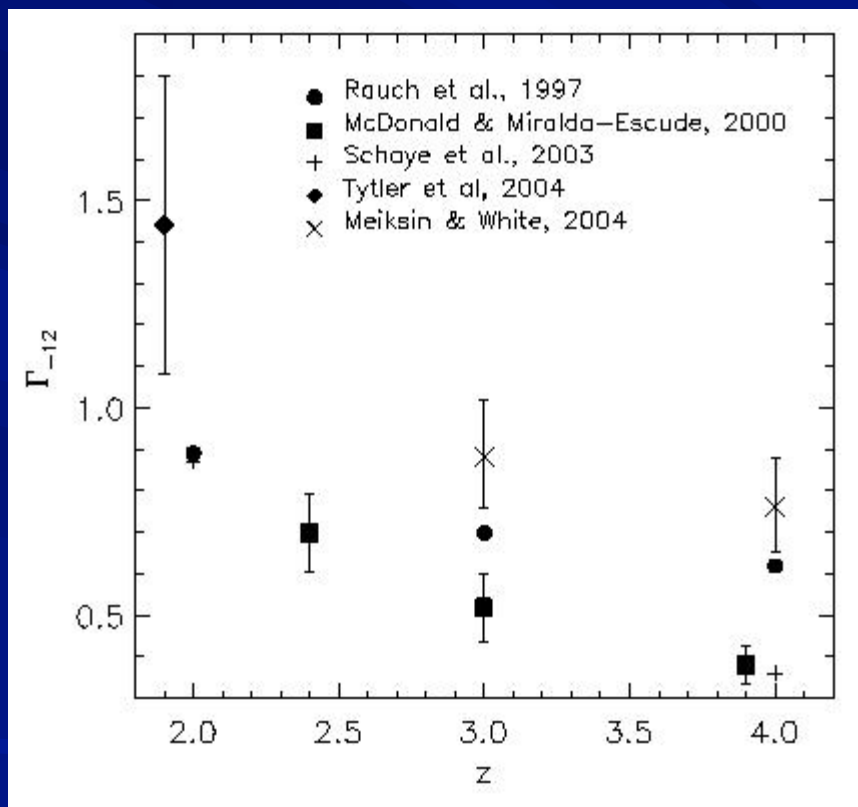
- Proximity effect (*e.g.* Scott *et al.* 2000)
- Lyman continuum emission from LBGs (*e.g.* Steidel *et al.* 2001)
- Modelling QSO population evolution (*e.g.* Haardt & Madau 1996)
- We use the Ly- α forest opacity to determine Γ_{HI} for $2 < z < 4$ with hydrodynamical simulations

Obtaining Γ_{HI} from simulations



- Hydrodynamical simulations of structure formation can be calibrated to reproduce popular parameters which influence the Ly α forest opacity ($\Omega_m, \Omega_b, h, \sigma_8, n, T_{\text{IGM}}$).
- Immerse box in a uniform UV background, keep its intensity as a free parameter.
- Rescale artificial spectra in post-processing to reproduce observed Ly- α forest opacity (e.g. Rauch *et al.* 1997, Theuns *et al.* 1998)

Estimates of Γ_{HI} from simulations



$$\Gamma_{-12} = \frac{\Gamma_{\text{HI}}}{10^{-12} \text{ s}^{-1}}$$

Lyman- α forest opacity

The Fluctuating Gunn Peterson Approximation:

- Assume photoionization equilibrium and an effective equation of state for low density gas, $T = T_0 \Delta^{\gamma-1}$ (Hui & Gnedin, 1997)

$$\tau = \tau_0 \frac{(1+z)^6 (\Omega_b h^2)^2}{T^{0.7} H(z) \Gamma_{-12}(z)} \Delta^\beta$$
$$\Rightarrow \Gamma_{-12} \propto \Omega_b^2 h^3 T^{-0.7} \Omega_m^{-0.5} \quad z \geq 2$$

e.g. Rauch *et al.* 1997, McDonald & Miralda-Escudé 2000

Our Aims

- Investigate the dependence of Γ_{HI} on a wide range of physical and numerical parameters with state-of-the-art cosmological simulations (GADGET-2).

Why?

- Understand the physical processes which alter the value of Γ_{HI} inferred from simulations.
- Determine how Γ_{HI} scales with the simulation input parameters (e.g. T_{IGM} , Ω_{m} , σ_8); estimate the uncertainty on Γ_{HI} obtained from simulations.
- Which sources contribute to the UV background for $2 < z < 4$?

Fiducial Model Parameters

Cosmological parameters consistent with Spergel *et al.* (2003)

$$\begin{aligned}\Omega_m &= 0.26 \pm 0.04 \\ \Omega_b h^2 &= 0.024 \pm 0.001 \\ \sigma_8 &= 0.85 \pm 0.05 \\ h &= 0.72 \pm 0.04 \\ n &= 0.95\end{aligned}$$

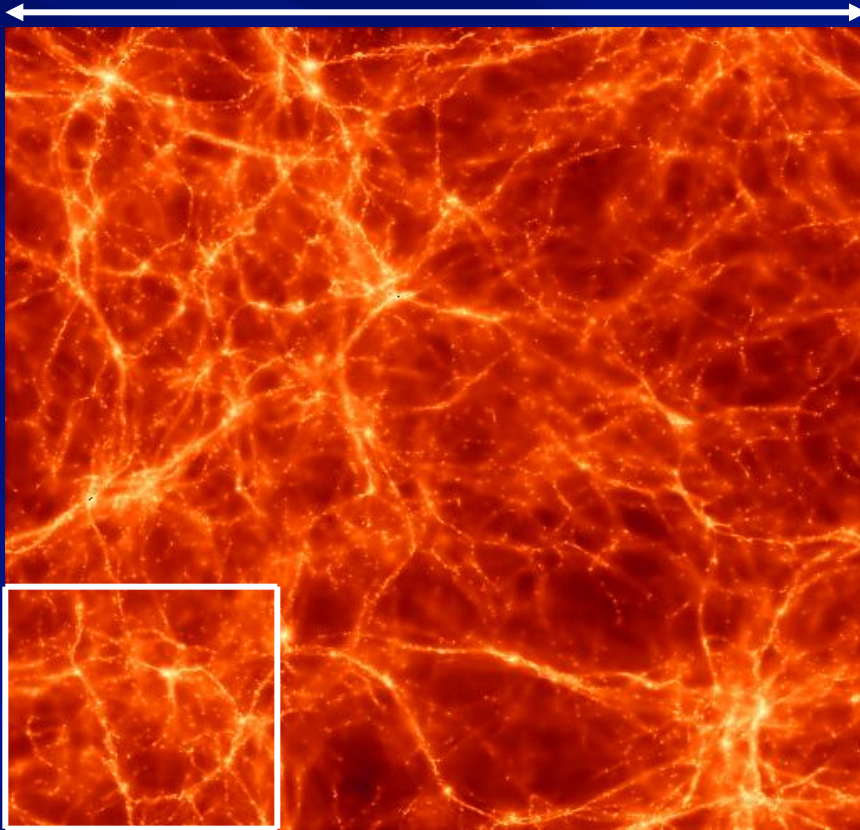
Astrophysical parameters at $z = [2, 3, 4]$

$$\begin{aligned}T_{\text{IGM}} &= [11200, 17800, 12500] \pm 5000 \text{ K} \\ \gamma &= 1.3 \pm 0.3 \\ &(\text{Schaye } et al. 2000)\end{aligned}$$

$$\begin{aligned}\tau_{\text{eff}} &= [0.130 \pm 0.021, 0.362 \pm 0.036, 0.805 \pm 0.070] \\ &(\text{Schaye } et al. 2003)\end{aligned}$$

Resolution and box size

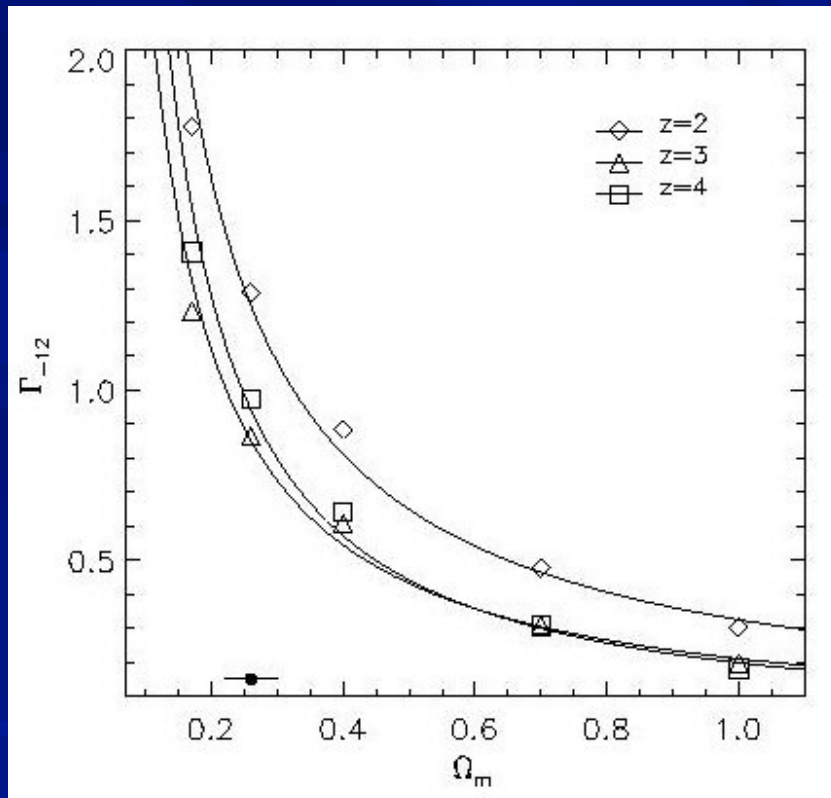
30 Mpc/h (Bolton *et al.* 2005)



- Large volume required to include long wavelength perturbations and provide an adequate sample of the Universe.
- High resolution required to resolve small haloes.
- Minimum box size and resolution of 30 Mpc/h and 400^3 gas particles required for marginal convergence of Γ_{HI} .

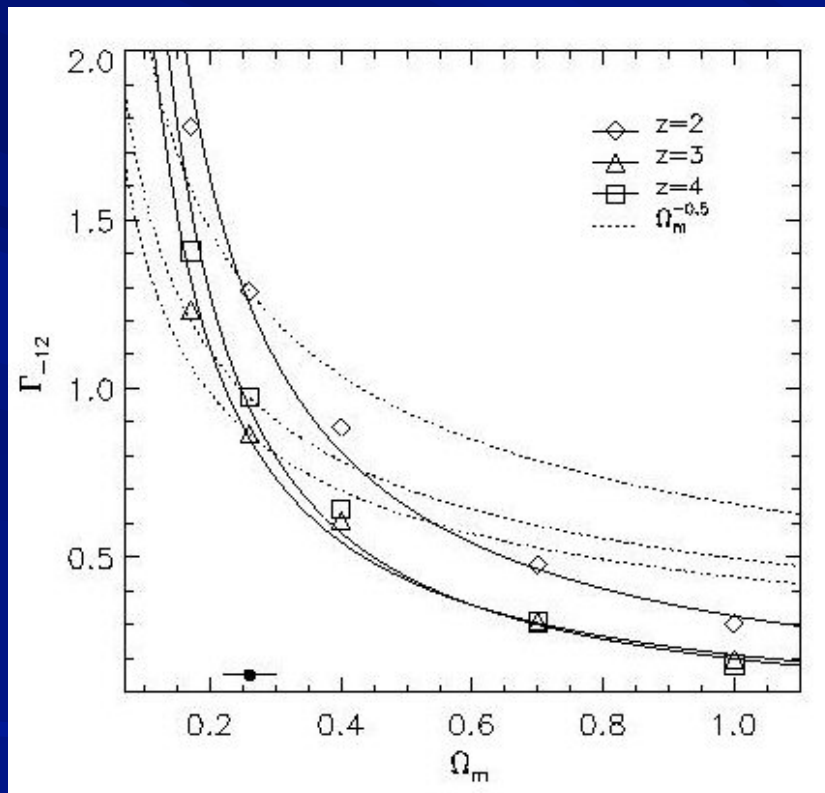
10 Mpc/h (Rauch *et al.* 1997)

Scaling with Ω_m



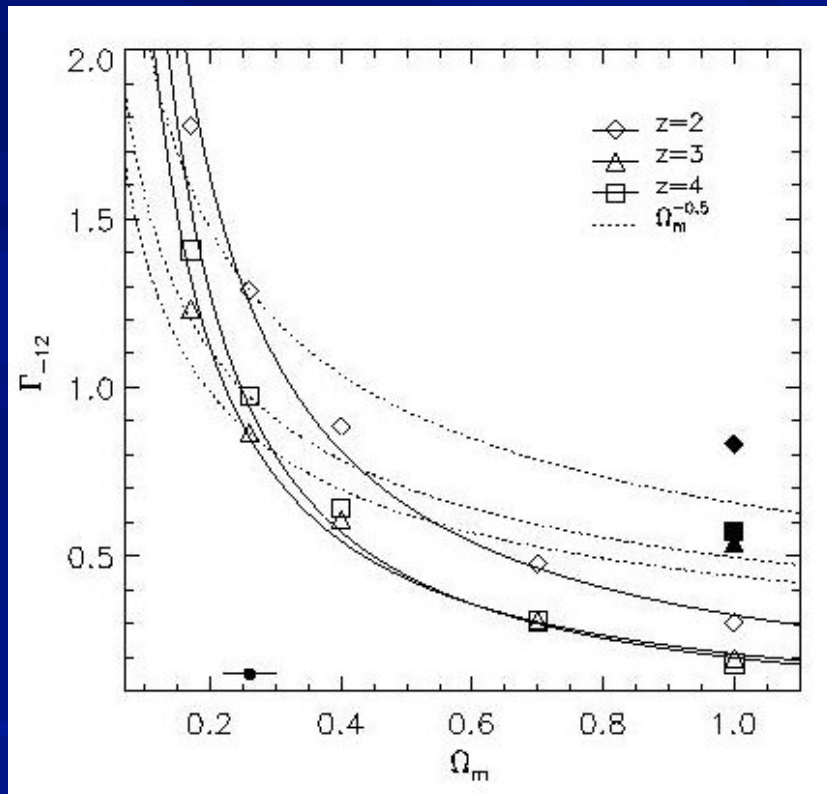
- Lower Ω_m models have less gas in haloes, so a larger Γ_{-12} is required to match the observed opacity.

Scaling with Ω_m



- Significant departure from the predicted scaling of Γ_{-12} with $\Omega_m^{-0.5}$ when normalised to the fiducial model

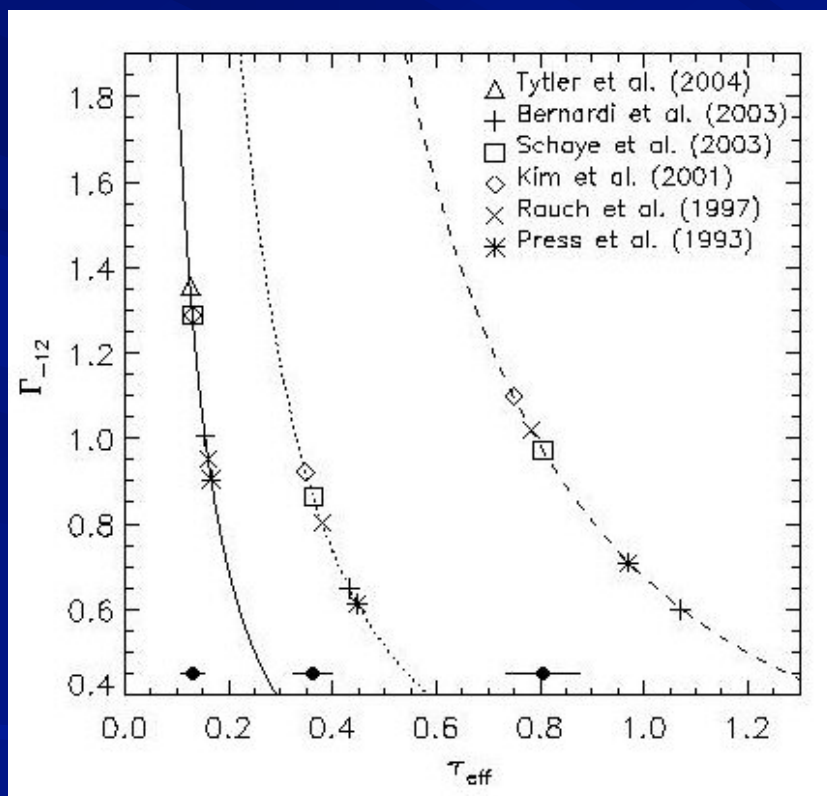
Scaling with Ω_m



- Extra simulation with $\Omega_m=1$; power spectrum normalised to have same fluctuation amplitude as the fiducial model at 30 kms⁻¹ scale.
- The *r.m.s* fluctuation amplitude at a fixed velocity scale is more relevant than the geometrical scaling of Γ_{HI} with $\Omega_m^{-0.5}$ from the Hubble parameter.

JSB, Haehnelt, Viel & Springel, 2005

Scaling with τ_{eff}



- We must assume a value of τ_{eff} to rescale the simulated spectra opacity and hence infer Γ_{HI}

- Systematic uncertainties stemming from the continuum fitting produce a wide range of estimates.

- A small change in τ_{eff} can have a dramatic effect on Γ_{-12}

JSB, Haehnelt, Viel & Springel, 2005

Uncertainties (%) and Results

Parameter	z=2.0	z=3.0	z=4.0
T	+50 -22	+23 -14	+35 -18
Ω_m	+18 -13	+19 -14	+21 -15
τ_{eff}	+29 -19	+18 -14	+17 -13
Numerical	± 10	± 10	± 10
γ	± 1	+7 -9	+12 -13
$\Omega_b h^2$	+9 -8	+9 -8	+9 -8
σ_8	+6 -5	± 6	+8 -7
h	± 6	± 6	± 6
Total	+62 -36	+39 -30	+49 -34

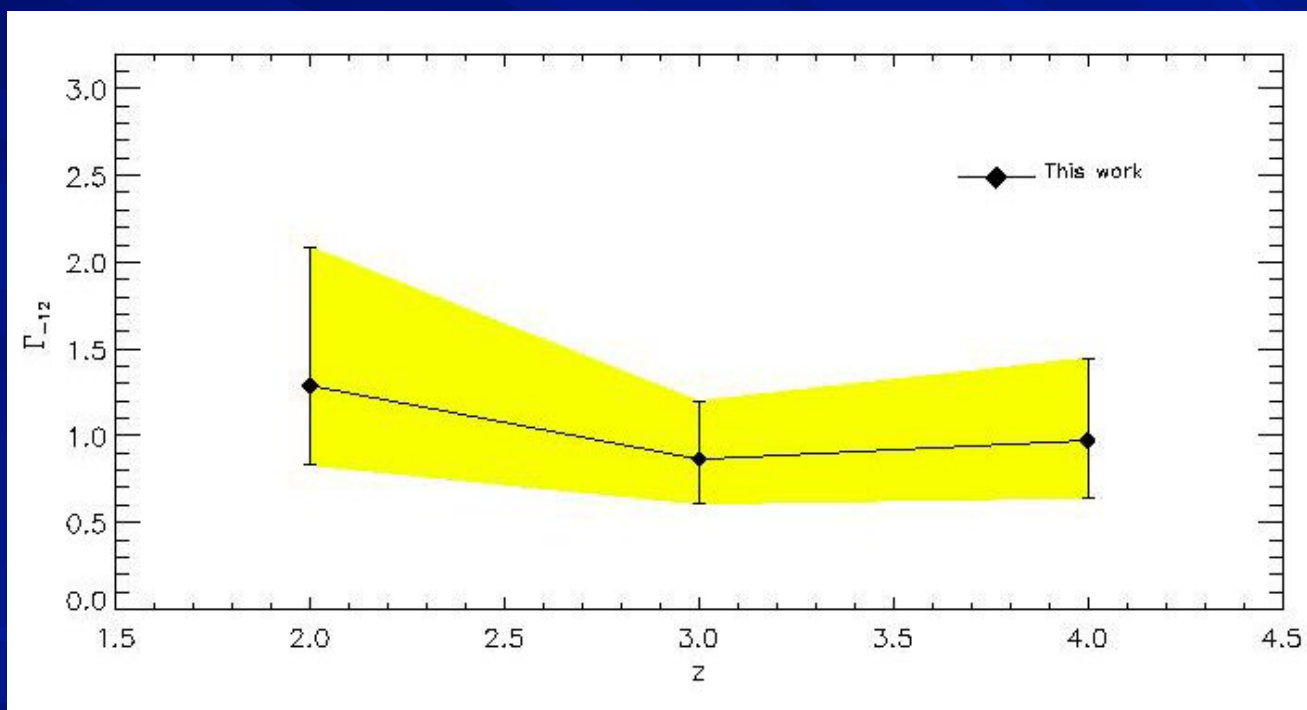
Final values

$$\Gamma_{-12} = [1.3 \pm_{0.5}^{0.8}, 0.9 \pm 0.3, 1.0 \pm_{0.3}^{0.5}]$$

at $z = [2, 3, 4]$

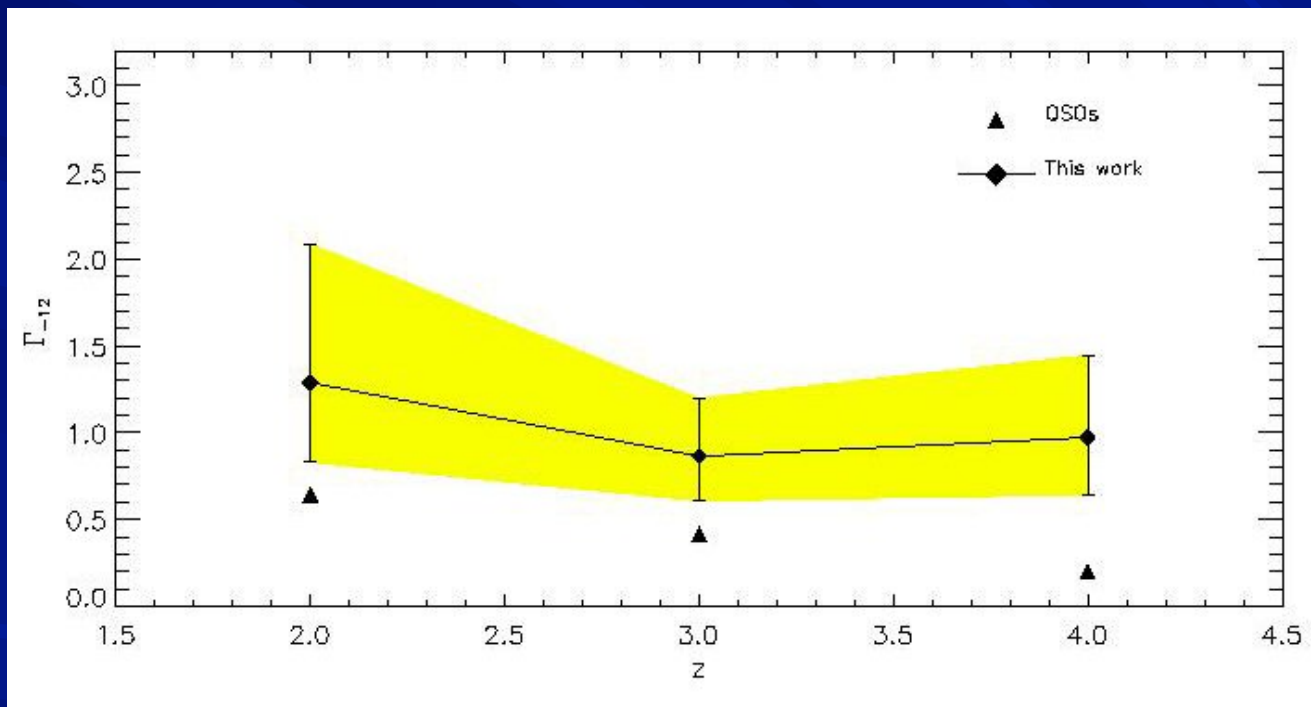
Comparison to other observations

Our results with uncertainties (Bolton *et al.* 2005)



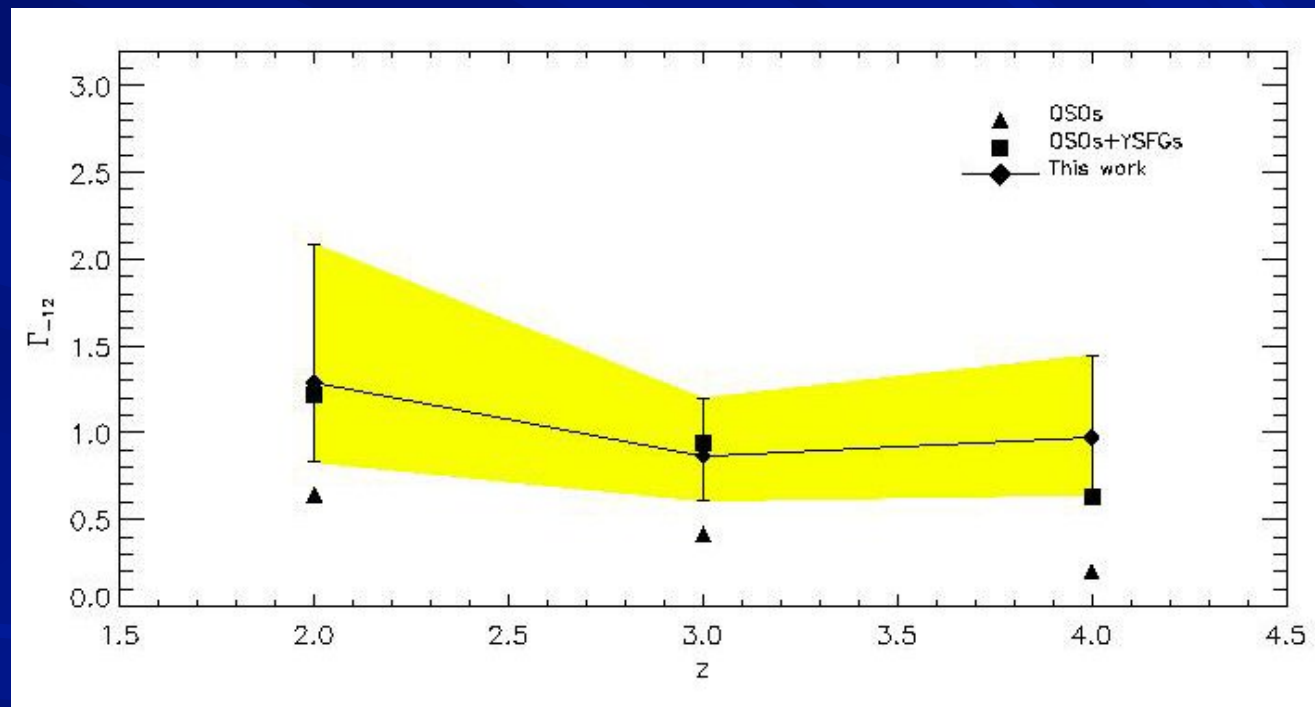
Comparison to other observations

Rates from QSOs (Boyle *et al.* 2000) + IGM re-emission
(Madau, Haardt & Rees 1999, updated)



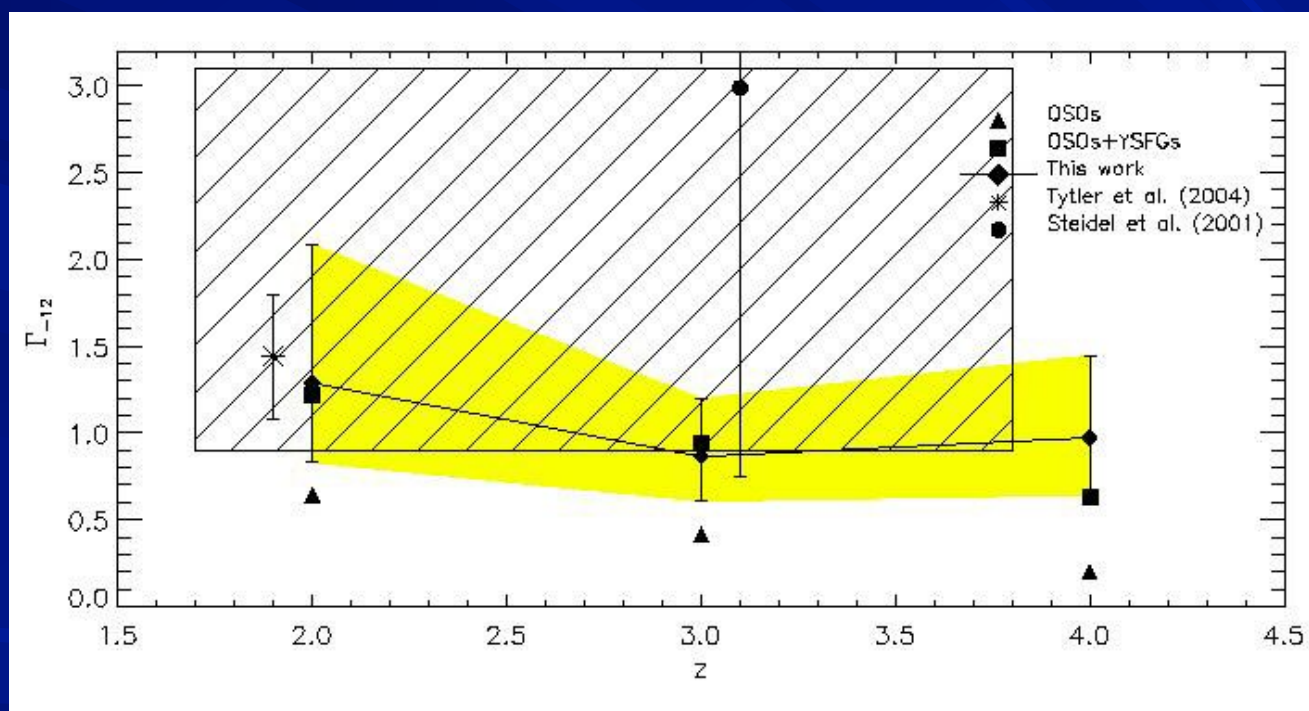
Comparison to other observations

Rates from galaxies (Bruzual & Charlot model), QSOs+IGM re-emission (Madau, Haardt & Rees 1999, updated)



Comparison to other observations

Proximity effect (Scott *et al.* 2000) and emission from Lyman-break galaxies (Steidel *et al.* 2001)



JSB, Haehnelt, Viel & Springel, 2005

Conclusions

- Our data are consistent with a UV background with a substantial contribution from galaxies, and agree with other observational estimates for the metagalactic hydrogen ionization rate.
- The thermal state of the IGM is the biggest uncertainty when determining the ionization rate.

Bolton *et al.*, 2005, MNRAS, 357, 1178